

LAB INSTRUCTION SHEET

Unit 3, User & Group Account Management

Duration: 2 hours | Mode: Individual | Environment: Linux VM (Ubuntu 22.04)

Learning Outcomes

By the end of this lab you should be able to:

- Create, modify, and delete user and group accounts using Linux command-line tools
- Configure password policies and account expiry using `passwd`, `chage`, and PAM
- Interpret the `/etc/passwd`, `/etc/shadow`, and `/etc/group` files
- Verify user and group membership using `id`, `groups`, and `getent`
- Apply the principle of least privilege when designing an account structure

Prerequisites

Before you begin:

You need a working Ubuntu 22.04 VM with `sudo` access. Confirm you can log in and run: `sudo whoami` (expected output: `root`). All tasks require a terminal session.

Background

In Linux, every process and file is owned by a user and a group. Proper account management is the foundation of the access control stack — if accounts are poorly managed (orphaned accounts, weak passwords, excessive group membership), all other security controls become weaker.

Key files you will work with:

File	Purpose
<code>/etc/passwd</code>	One line per user: username, UID, GID, home dir, shell (no passwords)
<code>/etc/shadow</code>	Hashed passwords + expiry metadata (readable only by root)
<code>/etc/group</code>	Group definitions: group name, GID, member list
<code>/etc/gshadow</code>	Group passwords and admin list (rarely used directly)

Task 1 — Create Users and Groups

01 Create the required users and groups for a small development team

1.1 Create Groups

Create the following groups on your system:

- developers
- devops
- auditors

```
# Create three groups
sudo groupadd developers
sudo groupadd devops
sudo groupadd auditors

# Verify groups were created
getent group developers devops auditors
```

Observation

Run: `grep -E 'developers|devops|auditors' /etc/group`

Record the GIDs assigned to each group:

Group	GID	Initial Members
developers		
devops		
auditors		

1.2 Create Users

Create four users with appropriate settings:

Username	Full Name	Primary Group	Secondary Groups
alice	Alice Nkurunziza	developers	devops
bob	Bob Mugisha	developers	—
charlie	Charlie Uwase	devops	auditors
diana	Diana Habimana	auditors	—

```
# Create users with home directories and specified primary group
sudo useradd -m -c "Alice Nkurunziza" -g developers -G devops alice
sudo useradd -m -c "Bob Mugisha" -g developers bob
sudo useradd -m -c "Charlie Uwase" -g devops -G auditors charlie
sudo useradd -m -c "Diana Habimana" -g auditors diana

# Set initial passwords (use Lab2025! for this lab)
echo 'alice:Lab2025!' | sudo chpasswd
echo 'bob:Lab2025!' | sudo chpasswd
echo 'charlie:Lab2025!' | sudo chpasswd
echo 'diana:Lab2025!' | sudo chpasswd
```

Verify Users Were Created

```
# Check each user's identity
id alice
```

```
id bob
id charlie
id diana

# View /etc/passwd entries
grep -E 'alice|bob|charlie|diana' /etc/passwd
```

Questions 1.2

1. What is the UID assigned to alice? _____ What does the UID tell you about user privilege?
2. What is the difference between the primary group and secondary groups in the output of id?
3. Where is alice's home directory? What permissions does it have? (Run: ls -la /home/)

Task 2 — Modify Accounts and Group Membership

02 Use usermod and groupmod to update account properties

2.1 Modify User Attributes

```
# Change alice's login shell to bash (if not already set)
sudo usermod -s /bin/bash alice

# Add bob to the devops group (without removing his existing groups)
sudo usermod -aG devops bob

# Lock charlie's account temporarily
sudo usermod -L charlie

# Unlock charlie's account
sudo usermod -U charlie

# Verify changes
id bob
grep charlie /etc/shadow # look for ! prefix indicating lock
```

Warning: Using `usermod -G` (without `-a`) REPLACES all secondary groups. Always use `-aG` to ADD a group.

2.2 Manage Group Membership

```
# Add diana to the developers group
sudo gpasswd -a diana developers

# Remove bob from devops group
sudo gpasswd -d bob devops

# Verify
getent group developers
getent group devops
```

Questions 2.2

4. After running `'sudo usermod -L charlie'`, what change do you see in `/etc/shadow` for charlie's entry?
5. What is the risk of using `'usermod -G devops bob'` instead of `'usermod -aG devops bob'`?

Task 3 — Password Policy Configuration

03

Configure password ageing and expiry using chage and passwd

3.1 View Password Policy

```
# View alice's current password ageing info
sudo chage -l alice

# View shadow entry directly
sudo grep alice /etc/shadow
```

3.2 Set Password Policy for alice

```
# Set password policy:
#   Min age: 1 day      (cannot change more than once per day)
#   Max age: 90 days   (must change every 90 days)
#   Warning: 7 days    (warn 7 days before expiry)
#   Inactive: 14 days  (disable account 14 days after expiry)
sudo chage -m 1 -M 90 -W 7 -I 14 alice

# Verify
sudo chage -l alice
```

3.3 Force Password Change on Next Login

```
# Expire bob's password immediately — forces change on next login
sudo passwd -e bob

# Verify
sudo chage -l bob
```

Observation Table — Fill in after running chage -l alice

chage Field	Value for alice
Last password change	
Password expires	
Password inactive	
Account expires	
Minimum days between changes	

Maximum days between changes	
Number of days of warning	

Task 4 — Reading `/etc/passwd` and `/etc/shadow`

04 Interpret the fields in `/etc/passwd` and `/etc/shadow` for your users

Run the following command and paste alice's `/etc/passwd` entry below:

```
grep alice /etc/passwd
```

`/etc/passwd` format:

username : password_placeholder : UID : GID : GECOS : home_directory : login_shell

Fill in the fields for alice:

Field	Value for alice
Username	
Password placeholder (field 2)	
UID (field 3)	
GID (field 4)	
GECOS / Comment (field 5)	
Home directory	
Login shell	

Now examine the shadow entry:

```
sudo grep alice /etc/shadow
```

The second field is the hashed password. What hashing algorithm prefix do you see?

Task 5 — Delete an Account Safely

05 Remove a user account while preserving or archiving their files

Before deletion, always check what files the user owns and decide what to do with them.

```
# Find all files owned by bob
find / -user bob 2>/dev/null

# Archive bob's home directory before deletion
sudo tar -czf /root/bob_backup_$(date +%Y%m%d).tar.gz /home/bob

# Delete bob's account BUT keep home directory intact
sudo userdel bob
ls /home/bob      # home still exists

# Delete bob's account AND remove home directory
sudo userdel -r bob    # only run this AFTER archiving!

# Verify deletion
id bob 2>&l          # should say 'no such user'
ls /home/           # bob's home should be gone
```

Best practice: Always archive user data before deletion. In production, disable the account first (usermod -L), wait for audit clearance, then delete.

Questions 5

6. What happens to files owned by a deleted user if you do NOT use the -r flag? What UID shows in ls -la for those files?
7. Why is it recommended to disable an account before deleting it in a production environment?

Reflection & Discussion Questions

Discussion:

Consider these questions individually, then discuss with your group.

8. You discover a user account with UID 0 that is not root. What security concern does this raise and what would you do?
9. A colleague suggests creating shared accounts (e.g., one 'developer' login for the whole team) to simplify administration. What risks does this introduce? What is the better approach?
10. How does password ageing (chage settings) relate to the principle of defence in depth? What would happen if all passwords had max age = 9999 (never expire)?
11. Compare managing users via CLI commands vs. a GUI tool like Active Directory Users and Computers. In what scenarios is each approach preferred?

Submission Checklist

- All four users created with correct group assignments (verified with id)
- Password policy applied to alice (verified with chage -l alice)
- Observation table for /etc/passwd fields completed
- chage output table filled in
- All numbered questions answered
- bob's account deleted using the safe archival procedure

Submission:

Submit this completed sheet plus a screenshot of: (1) id alice output, (2) chage -l alice output, (3) getent group developers output.